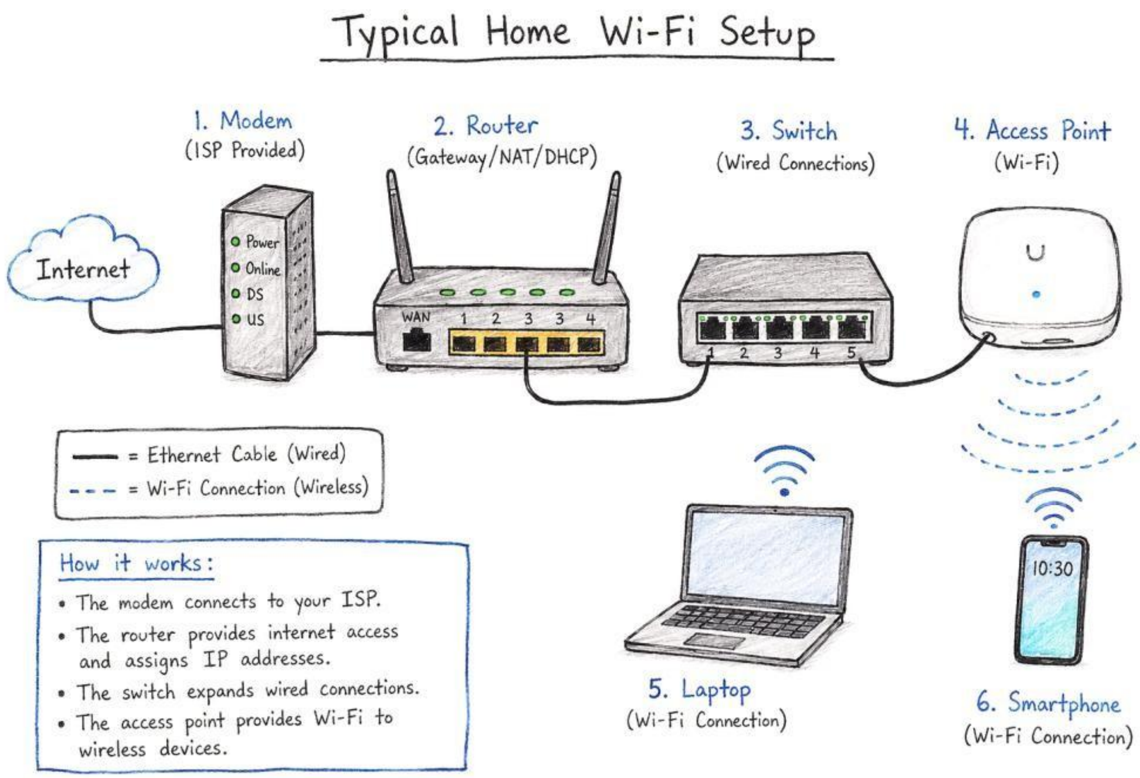
Introduction to Networking Hardware Theory

1.Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet

Device Function Example

Router Shares Internet Home

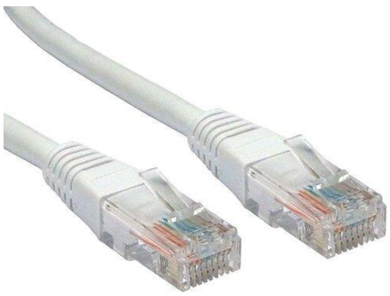
Internet

Switch Connect wired devices Home Wi-Fi

Access Point Gives Wi-Fi access School computer lab Network Cable

Connects devices with a wire PC to router

3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.



i. Ethernet Cable :- Used to connect computer, routers, and switches in homes and offices.

ii. Coaxial cable :- Used for cable tv and broadband internet connection.

iii. Fiber Optic Cable :- Used for high- speed internet and long- distance data transmission.

4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

= Feature	Router	Switch
Use	Shares internet	Connects devices.
Location	Home	Office/School
Data	Between networks	Within one network
Ports	Few ports	Many ports.